Day 6: Introduction to Network Devices Theory (1 hour): Overview of switches, routers, hubs, firewalls, and access points. Practical (1 hour): identify ports and functions of network devices in a setup.Task: Create a table comparing network devices and their features.

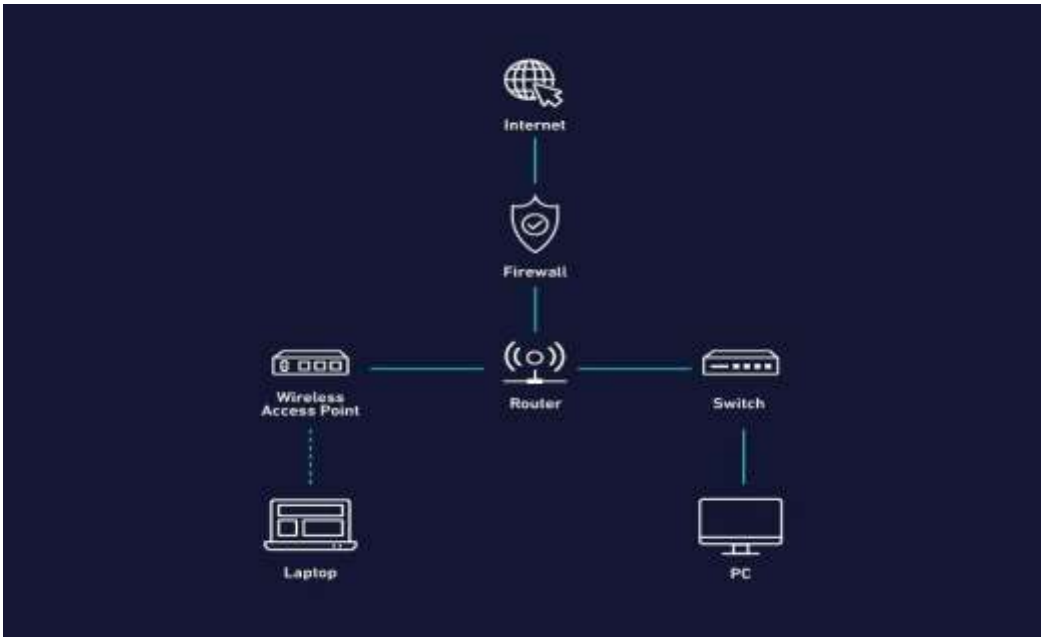
Tasks

1.Create a table listing the main functions and typical ports used by switches, routers, hubs, firewalls, and access points in a home WiFi setup.

ANS:

Network Devices Comparison Table

Device	Main Function	Typical Ports
Switch	Connects multiple devices in a local network.	5, 8, 16, 24 Ethernet ports
Router	Connects the home network to the internet.	WAN port, LAN ports
Hub	Connects multiple devices and sends data to all connected ports.	4, 8, 16 Ethernet ports
Firewall	Protects the network from unwanted and harmful traffic.	WAN, LAN, management ports
Access Point	Provides Wi-Fi connection to mobile phones and laptops.	Ethernet port, PoE port



2.Given images or diagrams of network devices, label each device and mark the ports (Ethernet, console, power, antenna) for at least three devices.

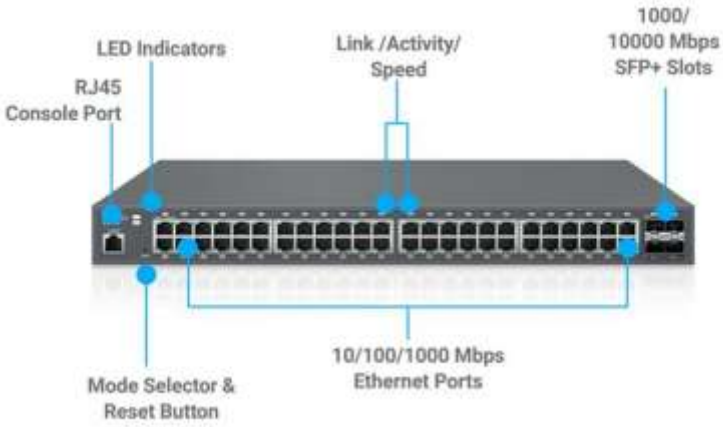
ANS:

1.Router



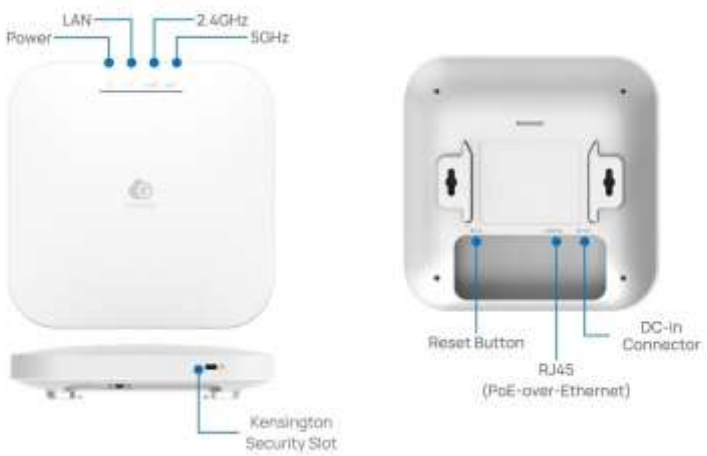
1. WAN Port – Connects the router to the modem/Internet.
2. LAN Ethernet Ports – Connect computers and other wired devices.
3. Power Port – Supplies electrical power.
4. Antennas – Transmit and receive Wi-Fi signals.
5. Reset Button – Restores factory settings.

2. Network Switch



1. Ethernet Ports – Connect multiple computers and network devices.
2. Console Port – Used to configure and manage the switch.
3. Power Port – Supplies electrical power.
4. Status LEDs – Show connection and network activity.

3. Wireless Access Point (AP)



1. Ethernet Port – Connects the access point to a router or switch.
2. Power Port – Supplies electrical power (if provided).
3. Antennas – Send and receive wireless signals.
4. Reset Button – Restores default settings.

3.Pick a popular app you use daily (like Instagram, Zomato, or Paytm) and describe in 3-4 lines how at least two different network devices (from switches, routers, firewalls, access points) play a role when you use that app.

ANS:

1. When I use Instagram on my mobile phone, the **Wi-Fi Access Point** connects my phone to the wireless network.
2. The **Router** forwards my data through the internet to Instagram's servers.
3. A **Firewall** helps protect the network by blocking unauthorized or suspicious traffic.
4. Network **Switches** help transfer data between devices within the network and connect servers.

4.Suppose your college WiFi is slow and sometimes disconnects. Based on the features of access points, switches, and routers, suggest one possible reason for the issue and how upgrading one device could help.

Hint: Think about device capacity, coverage, and traffic handling.

ANS: One possible reason for slow college Wi-Fi is that too many students are connected to the same Wi-Fi access point, exceeding its capacity. This can cause slow internet speed and frequent disconnections.

Solution: The college should upgrade its existing Wi-Fi access points to modern, high-capacity access points that support more users and provide better coverage.